

Task 1: DNA Sequence Analysis

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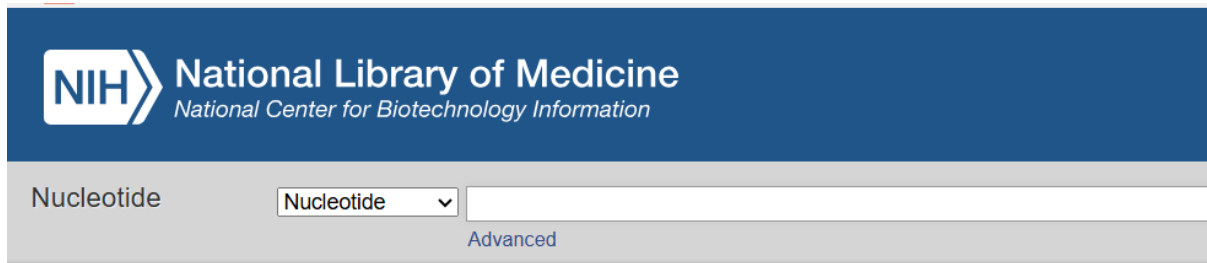
Student ID: CA/DF1/110014

Domain: Bioinformatics

Step 1. Retrieve the sequence

Search the accession **AY604160.1** in NCBI Nucleotide using:

Enter **AY604160.1** in the search box to open the sequence record. NCBI nucleotide records can be retrieved directly using accession numbers.



[FASTA](#) ▾

Pan paniscus Alu Sq repeat region

GenBank: AY604160.1

[GenBank](#) [Graphics](#)

```
>AY604160.1 Pan paniscus Alu Sq repeat region
CATGGCCATCACAACAGAAGCTCTAAGACGTTAAACAAGCAACATGTCCTAAGTGAAGTGTCCAGTTCA
ATTTTATTTATTTATTTTCTTGAGATGGAGTCTCACTCTTGTTGCCAGGCTGGAGTGCAGTGGTGCA
ATATTGGCTCACTGCAACCTCTGTCTCCAGGTTCAAGTGATTCTCCTGCCTCAGCATCCTGAGTAGCTG
GGATTACAGGTGCCAACCACACACCTGGATAACTTTTTGTATTTTAGTAGAGACGGGGTTTCACCAT
GTTGCCAGGCTGGTCTTGAACCTCTGACCTCAGGTGATCCGCCCGCCTCGGCCTCCCAAAGTGCTGGGA
TTACAGGTGCGAGCCACCGTGCCCGGCTCATTAAAGTATTTCCATAATGATTTTCAAGAAGACA
CTGAGAACATGTGAATGAAACCCACGCATGACACAACACTGGGAGACAGAAACAGTATTTTCATCTGCCAA
ACAGGG
```

Step 2. Download the FASTA sequence

1. Open the record.
2. Click **Send to** → **File**.
3. Select **FASTA**.
4. Save the file.

Step 3. Run BLAST

Go to: NCBI_BLAST

Choose: **Nucleotide BLAST (BLASTn)**

Paste the FASTA sequence.

Keep the default settings.

Click **BLAST**.

The screenshot shows the NCBI BLAST search interface. The 'Enter Query Sequence' section contains a text box with a FASTA sequence:

```
ATAATGATTTTCAAGAAGACA
CTGAGAACATGTGAATGAAACCCACGCATGACACAACACTGGGAGACAG
AAACAGTATTTCATCTGCCAA
ACAGGG
```

. Below this is a 'Job Title' field and a checkbox for 'Align two or more sequences'. The 'Choose Search Set' section shows the 'Database' dropdown set to 'Nucleotide collection (nr/nt)'. The 'Organism' field is empty. The 'Exclude' section has checkboxes for 'Models (XM/XP)' and 'Uncultured/environmental sample sequences'. The 'Limit to' section has a checkbox for 'Sequences from type material'. The 'Entrez® Query' field is empty. The 'Program Selection' section shows 'Optimize for' with radio buttons for 'Highly similar sequences (megablast)', 'More dissimilar sequences (discontiguous megablast)', and 'Somewhat similar sequences (blastn)'. The 'BLAST' button is highlighted in blue. A note at the bottom states: 'Note: Parameter values that differ from the default are highlighted in yellow and marked with + sign'.

Step 4. Record these values

From the first (best) alignment note:

Parameter	Description
Query Coverage	100%
Percent Identity	100%
E-value	0.0
Max Score	954

[Edit Search](#) [Save Search](#) [Search Summary](#) ▼

[How to read this report?](#) [BLAST Help Videos](#) [Back to Traditional Results P](#)

Job Title	Nucleotide Sequence
RID	3Z0FAEMR016 Search expires on 06-28 15:55 pm Download All ▼
Program	BLASTN Citation ▼
Database	nt See details ▼
Query ID	Icl Query_6252891
Description	None
Molecule type	dna
Query Length	496
Other reports	Distance tree of results MSA viewer ?

Filter Results

Organism only top 20 will appear ☐ **excl**

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity to **E value** to **Query Coverage** to

[Filter](#) [Reset](#)

Descriptions [Graphic Summary](#) [Alignments](#) [Taxonomy](#)

Sequences producing significant alignments

[Download](#) ▼ [Select columns](#) ▼ Show [?](#)

☒ select all 10 sequences selected

[GenBank](#) [Graphics](#) [Distance tree of results](#) [MSA View](#)

	Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
☒	Homo sapiens Alu repeat region	Homo sapiens	954	954	100%	0.0	100.00%	496	AY604158
☒	Gorilla gorilla Alu repeat region	Gorilla gorilla	931	931	100%	0.0	99.19%	497	AY604157
☒	Homo sapiens chromosome 8 clone CH17-241F1 . complete sequence	Homo sapiens	896	9738	100%	0.0	97.98%	243888	AC270296
☒	Homo sapiens chromosome 8 . clone RP11-212F11 . complete sequence	Homo sapiens	896	4599	100%	0.0	97.98%	167547	AC112673
☒	Pan paniscus Alu Sq repeat region	Pan paniscus	539	539	62%	1e-147	97.07%	307	AY604164
☒	Eukaryotic synthetic construct chromosome 13	eukaryotic synthetic construct	421	1.067e+05	64%	2e-112	92.96%	96089878	CP034516
☒	Eukaryotic synthetic construct chromosome 13	eukaryotic synthetic construct	421	1.067e+05	64%	2e-112	92.96%	96089878	CP034491
☒	Homo sapiens chromosome 2 clone CH17-218L2 . complete sequence	Homo sapiens	421	21183	63%	2e-112	92.68%	233881	AC277931
☒	Human DNA sequence from clone RP11-169O17 on chromosome 13 . complete sequence	Homo sapiens	421	8190	62%	2e-112	92.96%	190765	AL359763
☒	Homo sapiens chromosome 5 clone CH17-458O9 . complete sequence	Homo sapiens	417	35059	62%	3e-111	91.58%	209864	AC277898